

# Day 15 — AI Email Classification Workflow

n8n LLM Integration | Basic LLM Chain + Switch Routing

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## Intern Information

| Field                  | Detail             |
|------------------------|--------------------|
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## Prompt Used (Basic LLM Chain)

## ## AI Email Classification Prompt

You are an email classification system. Your ONLY job is to classify the email into exactly ONE category.

Allowed categories:

Sales  
Support  
Complaint  
Invoice  
Spam  
General

Follow these rules:

- SALES: The sender wants to buy something, asks about product/service prices, plans, packages, availability, or purchasing.
- SUPPORT: The sender needs help with an existing product/service, such as login problems, technical problems, account problems, or how-to questions.
- COMPLAINT: The sender is unhappy, dissatisfied, reports poor service, damaged products, delays, or a bad experience.
- INVOICE: The sender asks about an invoice, bill, receipt, payment, transaction, or billing document.
- SPAM: The email is suspicious, unsolicited, irrelevant, or clearly unwanted promotional content.
- GENERAL: Use this ONLY when the email does not clearly match any of the five categories above.

IMPORTANT:

- Choose the category based on the main purpose of the email.
- Do NOT choose General if the email clearly matches another category.
- Return ONLY ONE category name.
- Do not provide an explanation.
- Do not add punctuation.
- Do not write "Category:".

Email to classify:

{{ \$json.textPlain }}

# Classification Test Screenshots

Three separate executions of the workflow, each showing the Switch node routing a different email to a different category output.

## Classification 1 — Sales

The screenshot shows the n8n workflow editor for 'AI Email Classification'. The workflow consists of an 'Email Trigger (IMAP)' node, a 'Basic LLM Chain' node, and a 'Switch' node. The 'Switch' node is configured with five branches: 'Support', 'Sales', 'Complaint', 'Invoice', and 'Spam'. The 'Sales' branch is highlighted in green, indicating it is the active path. A notification box at the bottom right states 'Node executed successfully'. The logs section shows the following execution details:

| Step                 | Duration          | Start Time                   |
|----------------------|-------------------|------------------------------|
| Email Trigger (IMAP) | Success in 9.933s | Started 07:28:30.703, 19 Aug |
| Basic LLM Chain      | Success in 948ms  | Started 07:29:42.662, 19 Aug |
| Switch               | Success in 46ms   | Started 07:29:49.209, 19 Aug |

Switch output: "Sales" → routed to Sales branch

## Classification 2 — Spam

The screenshot shows the n8n workflow editor for 'AI Email Classification'. The workflow is the same as in the first screenshot, but the 'Spam' branch of the 'Switch' node is highlighted in green, indicating it is the active path. The logs section shows the following execution details:

| Step                 | Duration         | Start Time                   |
|----------------------|------------------|------------------------------|
| Email Trigger (IMAP) | Success in 647ms | Started 07:29:49.209, 19 Aug |
| Basic LLM Chain      | Success in 948ms | Started 07:29:49.209, 19 Aug |
| Switch               | Success in 20ms  | Started 07:29:49.209, 19 Aug |

The 'OUTPUT' section shows the following results:

| Output            | Value |
|-------------------|-------|
| Output 0          |       |
| Output 1          |       |
| Output 2          |       |
| Output 3          |       |
| Output 4 (1 item) | Spam  |
| Output 5          |       |

Switch output: "Spam" → routed to Spam branch (Output 4)

### Classification 3 — General

The screenshot displays the n8n workflow editor for a project named "AI Email Classification". The workflow is composed of several nodes: "Email Trigger (IMAP)", "Basic LLM Chain", "Group Chat Model", and a "Switch" node. The "Switch" node is configured with five branches: "Support", "Sales", "Complaints", "Service", and "General". Each branch is connected to a corresponding output node. The "General" branch is highlighted in green, indicating it is the active output. The "Output" tab shows "Output 5 (1 item)" with the text "General".

Logs: Success in 295ms

Switch Success in 7ms

OUTPUT

| Output 0 | Output 1 | Output 2 | Output 3 | Output 4 | Output 5 (1 item) |
|----------|----------|----------|----------|----------|-------------------|
|          |          |          |          |          | General           |

Switch output: "General" → routed to General branch (Output 5)